

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM 1 & 2		
SYSTEM:	Chemical and Volume Control		
TASK:	Demonstrate the Operability of the RWST and/or the BASTs		
TASK NUMBER:	N0040420201		
JPM NUMBER:	11-01 ILOT RO Admin A2		
ALTERNATE PATH:	☐	K/A NUMBER:	2.2.37
APPLICABILITY:		IMPORTANCE FACTOR:	
EO ☐	RO ☒	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:	Simulator - Perform		
REFERENCES:	S2.OP-ST.CVC-0010, Rev. 9 (checked 8-13-12) Salem Tech Specs		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	<u>13 minutes</u>		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	<u>N/A</u>		
Developed By:	G Gauding Instructor	Date:	8-4-12
Validated By:	Brennan SME or Instructor	Date:	8-24-12
Approved By:	Training Department	Date:	
Approved By:	Operations Department	Date:	
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:			DATE:

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SIMULATOR SETUP: **IC-231**

Reset IC for each JPM and ensure Plant Computer is returned to a neutral screen.

RWST levels -	CH I	41.4
	CH II	41.4
	CH III	41.4
	CH IV	41.5
BAST levels -	21 BAST	49.5
	22 BAST	49.5

INITIAL CONDITIONS:

- Salem Unit 2 is at 100% power.
- 21 BAT pump developed a leak that went unnoticed until the cross-connected BAST's reached their low level alarm setpoint.
- 21 BAT pump, and the leak, are isolated.
- BAST's remain cross-connected.
- Chemistry reports current RWST and BAST boron concentrations are:
 - 21 BAST – 6610 ppm
 - 22 BAST – 6610 ppm
 - RWST – 2350 ppm
- Off Normal and Off-Normal Tagged list review has been performed SAT.

INITIATING CUE:

You are the Reactor Operator. Perform S2.OP-ST.CVC-0010, Borated Water Sources.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

1. Complete Attachment 2 of S2.OP-ST.CVC-0010 per key and determine the Surveillance is SAT

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*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Provide marked up copy of S2.OP-ST.CVC-0010, Borated Water Sources.			
	2.1	IDENTIFY sections of this procedure that are <u>NOT</u> required to be performed.	Marks Section 5.2 N/A. Marks Attachment 3 (Modes 5-6) N/A or removes from procedure.		
	2.2	REVIEW Components "Off Normal and Off-Normal Tagged" List(s) for system and support system(s) associated with evolution to be performed in this procedure.	Determines from initial conditions that Components "Off Normal and Off-Normal Tagged" List(s) review has been performed SAT.		
	2.3	ENSURE calibration data for the instruments listed in Attachment 1, as applicable, is obtained.	Verifies calibration data for the instruments listed in Attachment 1 is recorded.		
	2.4	ENSURE current RWST and BAST concentrations are obtained by Chemistry.	Determines current RWST and BAST concentrations from initial conditions.		

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	3.0	Steps identified with dollar signs (\$) are those items required to meet Technical Specification acceptance criteria. Such steps, if not satisfactorily completed, may have reportability requirements and shall be brought to the immediate attention of the SM/CRS.	Reads and initials step.		
	5.1.1	COMPLETE Attachment 2 per the following: • RECORD Data. • RECORD Test Results by initialing SAT or UNSAT column using the stated Acceptance Criteria.	Completes Attachment 2 with the below data.		
*	Att. 2	RWST Level	Records RWST level with channel for which calibration data was obtained RWST levels are: CH I: 41.4 CH II: 41.4 CH III: 41.4 CH IV: 41.5 Initials SAT for RWST Level and inserts date.		

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*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	Att. 2	RWST Temp	Checks P-250 computer point T0650A and enters 70°F. Initials SAT for RWST Temp and inserts date.		
*	Att. 2	RWST Conc	Refers to initial conditions for RWST boron concentration and enters 2350 ppm. Initials SAT for RWST boron concentration and inserts date.		
*	Att. 2	21 BAST Vol <u>AND/OR</u> 22 BAST Vol	Records 21 and 22 BAST level indicated on control console from 2LI106 and 2LI102 respectively as 2LI106 – 49.5% 2LI102 - 49.5% Uses Tech Spec 3.1.2.6.a, 3.1.2.6.b, and Figure 3.1-2 and determines 21 BAST Vol <u>AND/OR</u> 22 BAST Vol is SAT. Initials SAT for BAST volume and inserts date.		

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*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	Att. 2	21 BAST Temp <u>AND/OR</u> 22 BAST Temp	Records 21 and 22 BAST temp indicated on control console from 2TI107 and 2TI103 respectively as: 2TI107 - 98.2 2TI103 - 97.1 Initials SAT for 21 BAST Temp <u>AND/OR</u> 22 BAST Temp and inserts date.		
*	Att. 2	21 BAST Conc <u>AND/OR</u> 22 BAST Conc	Refers to initial conditions for BAST boron concentrations and enters 6610 ppm for 21 BAST and 6610 ppm for 22 BAST. Uses Tech Spec 3.1.2.6.a, 3.1.2.6.b, and Figure 3.1-2 and determines 21 and 22 BAST boron conc is SAT. Initials and inserts date.		

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*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	5.3.1	This surveillance is considered satisfactory when Attachments 2 and 3 (as applicable) are complete with components listed meeting Acceptance Criteria stated in each Attachment.	Initials step 5.3.1		
	5.3.2	<u>OR</u> This surveillance is unsatisfactory. A. INITIATE Notification(s) to correct the unsatisfactory condition. B. RECORD Notification number(s) and reason for unsatisfactory completion on Attachment 4 in the Comments Section.	N/A's step 5.3.2		
	5.4.1	COMPLETE Attachment 4, Sections 1.0 and 2.0, <u>AND FORWARD</u> this procedure to the SM/CRS for review.	Completes Attachment 4, Sections 1.0 and 2.0, and returns procedure to JPM Evaluator.		
			JPM is complete when procedure is returned to evaluator.		

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- _____ 1. Task description and number, JPM description and number are identified.
- _____ 2. Knowledge and Abilities (K/A) references are included.
- _____ 3. Performance location specified. (in-plant, control room, or simulator)
- _____ 4. Initial setup conditions are identified.
- _____ 5. Initiating and terminating Cues are properly identified.
- _____ 6. Task standards identified and verified by SME review.
- _____ 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- _____ 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. _____ Date _____
- _____ 9. Pilot test the JPM:
 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: _____

Date: _____

SME/Instructor: _____

Date: _____

SME/Instructor: _____

Date: _____

INITIAL CONDITIONS:

- Salem Unit 2 is at 100% power.
- 21 BAT pump developed a leak that went unnoticed until the cross-connected BAST's reached their low level alarm setpoint.
- Chemistry reported current RWST and BAST boron concentrations last shift. No additions to those tanks have occurred since that report.
 - 21 BAST – 6610 ppm
 - 22 BAST – 6610 ppm
 - RWST – 2350 ppm
- Off Normal and Off-Normal Tagged list review has been performed SAT.

INITIATING CUE:

You are the Reactor Operator. Perform S2.OP-ST.CVC-0010, Borated Water Sources Data.